

Autonomous Mobile Surveying for Science Rovers using *In Situ* Distributed Remote Sensing

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Abstract—The search for signs of past or present life on planetary surfaces such as Mars is a high priority objective of international space agencies. At present and in the near future, the most promising means to conduct such searches employ robotic systems that operate from planetary orbits or on planetary surfaces. This paper presents a robotic system concept for mobile search and surveying on planetary surfaces. It employs an open-path absorption spectrometer with components distributed on a rover and terrain to measure biogenic gases in the near-surface atmosphere. Rover motion control algorithms are presented and demonstrated in simulation for surveying wide areas autonomously using the distributed spectrometer for *in situ* remote sensing.

I. INTRODUCTION

DETECTION of signatures that indicate the existence of biomarkers is pertinent to astrobiology investigations on Mars and to the eventual establishment of a human presence there. Space agencies will send landers and rovers to regions on planetary surfaces, where biomarker detection is likely, to conduct surveys in search of them. As a robotics application, we are developing an effective concept for performing mobile science surveys on planetary surfaces using a distributed spectrometer instrument system that employs a rover to achieve wide measurement coverage of a promising search area [1]. The general problem is directed wide-area survey and detection of biogenic gases (water vapor, methane, and others) in the ground-level atmosphere of Mars that may serve as indicators of past/present life. High-density concentrations of water vapor, methane, and other biogenic gases can be detected by absorption spectroscopy. The rover carries the spectrometer from place to place to acquire measurements of gas concentrations in the near-surface atmosphere via *in situ* remote sensing, with the overall goal to cover a designated area.

Remote sensor-based area coverage contrasts with more common area coverage problems for mobile robotic surveys that employ sensing devices that require close proximity to or contact with the measured phenomenon (e.g., ground penetrating radar [2], humanitarian de-mining with metal detectors [3], fluorescence imaging for organic molecule detection [4], etc). For those applications, the robot mobility system must physically cover most, if not all, of the terrain in

the survey area. Remote sensor-based methods are not applicable to all survey problems; when they are not the best solution, they are often excellent complements to contact-based methods, serving as an efficient means to survey wide areas to localize areas at which contact-based surveys should be done. Algorithms such as parallel line or parallel swath sweeps are commonly used to address such robotic contact-based area coverage problems [5].

This paper addresses aspects of the mobility and navigation problem associated with the instrument surveying functionality. The focus is on circular and spiral survey trajectories that are amenable to our system configuration and best exploit its distributed measurement coupling between a rover-based instrument and a retroreflector.

II. DISTRIBUTED MOBILE SPECTROMETER SYSTEM

The robotic system employed for this work consists of subsystems including a spectrometer instrument for acquiring scientific measurements, an acquisition and alignment subsystem used to point the spectrometer at a stationary retroreflector, and a mobility subsystem used to move the instrument spatially over terrain for surveying purposes. The instrument is an open-path absorption spectrometer that detects biogenic gases in the atmosphere. The spectrometer is mounted on a mobile robot and includes light sources in the form of near-infrared diode lasers, each tuned to the absorption frequency of a gas of interest. The lasers are aimed, in turn, at a stationary retroreflector that reflects the laser beam back to a detector on the instrument. If one of the gases is absorbed by reflected light impinging upon the detector, the instrument can measure its spectrum, and the presence and concentration of the gas in the atmosphere can be determined [6]. Laser emitter and detector components of the instrument and the retroreflector component are spatially distributed by a distance d , allowing an open path (of length $2d$) for the laser beam(s) through the atmosphere (Fig. 1). This laser beam path length is variable as d varies with rover position relative to the stationary retroreflector; for effective measurements it can vary from 1 to 100s of meters. While the spectrometer emitter-detector components are mounted on the rover, the retroreflector is rigidly attached atop a post that is either fixed to a lander or emplaced on the terrain; the post remains stationary during a survey at a position central to the survey area. A 360° retroreflector (such as those commonly used with theodolite systems by civil engineers for land surveys) is employed so that the rover can acquire measurements from any radial direction where the retroreflector is within line-of-sight.

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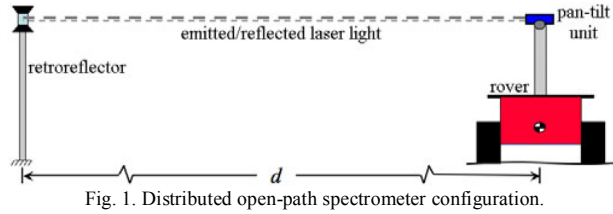


Fig. 1. Distributed open-path spectrometer configuration.

The acquisition and alignment subsystem employs a pan-tilt mechanism to aim the laser beam(s) at the passive retroreflector from a distance. To locate the retroreflector from afar, an acquisition sequence is used [7] in which a passive target in close proximity to the retroreflector is employed to facilitate its visual recognition from near and far distances. The rover uses a vision system to initially locate the target, processing its images to estimate the deviation from ideal alignment of the beam, and using that estimate to achieve fine alignment. On an early prototype [7], a circular target of 7.6 cm in diameter was used and the camera was a low-resolution monochrome unit with a 164x124 pixel array. With a 4.3 mm focal length lens, it had a field of view of 25° horizontal by 19° vertical. The pan-tilt mechanism resolution was sufficient to position the beam to within 1 cm of the target center at a range of 20 m. The latest prototype uses a color camera with a precision pan-tilt-zoom mechanism, and a target with a fixed color pattern [6].

The focus of this paper is the instrument mobility subsystem and its use for realizing the wide area surveying function of the system. The instrument mobility subsystem executes two types of survey trajectory as well as obstacle avoidance and biogenic gas source localization behaviors. Mobility algorithms provide a means to systematically acquire measurements covering an area to be surveyed in a discrete manner by transporting the instrument to multiple vantage points with respect to the retroreflector.

III. MOBILE SURVEY TRAJECTORIES

Rover survey trajectories are centered on the location of the retroreflector, which is assumed to be mounted on the spacecraft lander that delivered the rover to the planet surface. For the associated exploration scenario, open-path absorption spectrometer measurements would be acquired at discrete locations along the trajectory to achieve coverage of areas to be surveyed. Measurements along a line-of-sight between the rover-mounted laser emitter/detector and the distant retroreflector account for coverage of terrain below that line-of-sight. This is a key advantage of the open-path measurement configuration that permits a two-dimensional search over terrain using discrete linear measurements from a distance (similar to scanning laser rangefinders).

Various trajectories that circumnavigate the retroreflector at multiple distances away from it are possible. Circular or spiral survey trajectories are among the best candidates for our distributed spectrometer system configuration and measurement task. This is true when one considers

completeness of area coverage by *in situ* remote sensing and efficient use of rover resources (e.g., required energy for rover movements, traverse distance, etc) [8-10].

A. Concentric Circular Surveys

Beginning at a designated radial distance from the retroreflector, concentric circular trajectories of increasing radii can be followed to cover a designated area of interest, which we refer to as the *survey region*. While following a trajectory, the rover would move in arc-increments stopping periodically on the trajectory to acquire measurements along a ray between the rover-mounted laser bore-sight and the stationary retroreflector. By acquiring many measurements from different radial positions and distances, the rover samples the survey region in a discrete manner. We will refer to locations along a trajectory at which measurements should be acquired as *measurement nodes* (M-nodes).

The retroreflector location is known and considered to be the origin of an inertial coordinate system in which the survey region and task is defined. This survey coordinate system is oriented such that its ground-plane coordinate axes are aligned with true north and east. Rover pose during surveys is estimated relative to this coordinate system. An algorithm based on four key parameters is employed to command and execute a concentric circular trajectory for covering a given survey region. The survey trajectory and parameters are depicted in Fig. 2; the four parameters are:

- ρ : radial distance from the retroreflector at which the survey will start
- δ_c : separation distance between consecutive concentric circular trajectories
- s : arc length between consecutive M-nodes
- n : positive integer designating the outermost circle that encompasses the survey region.

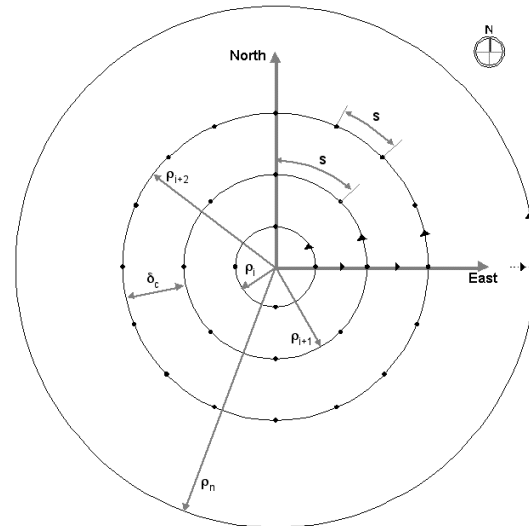


Fig. 2. Survey trajectory and parameters for concentric circular search.

Black dots in the figure represent M-nodes. The motion control algorithm for executing such surveys is described below where it is assumed that the rover is already within the

survey region and that the retroreflector is within line-of-sight from the rover. If the retroreflector is not within line-of-sight from a given M-node, no measurement is made. The survey completes when either the n^{th} trajectory is followed or a positive detection of biogenic gas is made.

Algorithm: Follow circular trajectory (ρ, δ_c, s, n)

1. Aim rover pointing system at retroreflector target; measure range, d_r , to retroreflector
2. Turn to orient rover along rover-retroreflector pointing direction
3. Drive toward first circular trajectory, until $d_r = \rho$
4. Execute circular survey
 - a. Orient rover tangent to circular trajectory ($\perp$ to rover-retroreflector pointing direction)
 - b. Execute survey circle i
 - i. Aim at retroreflector, acquire measurement(s)
 - ii. Move arc-length s to next M-node
 - iii. Repeat until circle i is complete
 - c. Orient away from retroreflector along rover-retroreflector pointing direction
 - d. Drive δ_c to next circle trajectory, until $d_r = \rho_i + \delta_c$
 - e. Next circle, $i = i + 1$
 - f. Repeat until circle n complete or gas detected.

The trajectory following algorithm must enable the rover to maneuver onto the trajectory from an off-trajectory pose. This applies at the start of a survey (steps 2 and 3 of the algorithm above). The same capability is needed when recovering from hazard avoidance maneuvers that cause the rover to deviate from the trajectory. The process of finding or recovering the trajectory from an off-trajectory position includes determining a close, free-space point (x_c, y_c) on the trajectory and navigating to it. This point can be taken to be at a distance from the rover position to the trajectory along a ray from the survey coordinate system origin through the rover position (i.e., along the rover-retroreflector pointing direction). From any such rover position, Cartesian (x_r, y_r) or polar (d_r, θ_r) , the ray through the origin is along the azimuth $\theta_r = \tan^{-1}(y_r/x_r)$; the rover must orient along θ_r and drive to (x_c, y_c) on the trajectory (or to distance $d_r = \rho$ from the origin). There are two cases to consider — the rover is inside or outside the desired circular trajectory, as illustrated in Fig. 3. Rover distance relative to the retroreflector, d_r , is measured by the instrument. If the rover is outside the circle of radius ρ that it must drive to, it must orient along θ_r toward the origin and drive inward for a distance, $(d_r - \rho)$; if it is inside that circle, it must orient away from the origin along θ_r and drive outward for a distance, $(\rho - d_r)$.

B. Spiral Surveys

As an alternative to circular trajectories, spiral trajectories of increasing radii can be followed to cover a designated survey region (Fig. 4). A similar trajectory following algorithm is used and is based on a similar set of four parameters. Parameters ρ and s are as defined earlier for circular trajectories while δ_s and n are defined as follows:

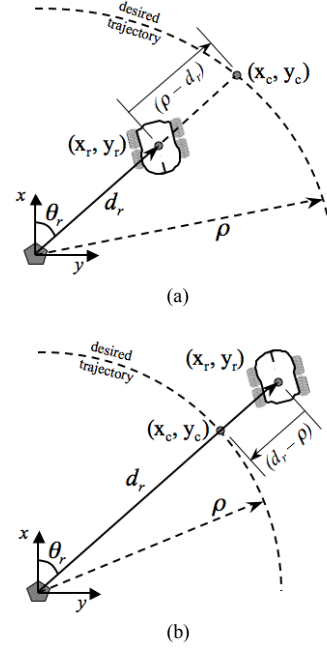


Fig. 3. Finding the trajectory from inside (a) or outside (b).

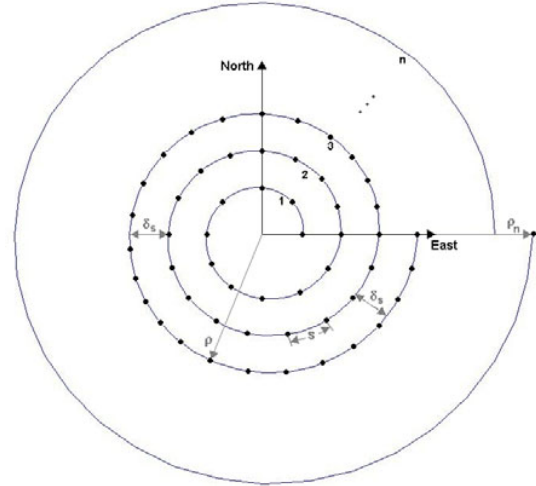


Fig. 4. Survey trajectory and parameters for linear spiral search.

- δ_s : separation distance between consecutive branches of the spiral trajectory
- n : positive integer designating the outermost spiral branch that encompasses the survey region.

We concentrate on linear spirals as survey trajectories that spiral outward from a given initial radius, ρ . For linear spirals, the radius ρ_s of the trajectory continually and gradually increases with distance traveled such that after every 2π radians consecutive branches of the spiral are separated by the fixed distance, δ_s . That is, (in polar coordinates) the radius increases linearly with the number of radians, θ , of the spiral. The linear spiral trajectory can be expressed in polar coordinate and Cartesian forms as follows

$$\begin{aligned}\rho_s(\theta) &= k\theta & (1) \\ x(\theta) &= k\theta \cos\theta, y(\theta) = k\theta \sin\theta & (2)\end{aligned}$$

where θ increases monotonically and k is a constant. For spiral branches, $\theta \in [2m\pi, 2(m+1)\pi]$; $m = \{0, 1, 2, \dots, n-1\}$. To achieve a desired separation distance, δ_s , between each successive spiral branch it can be shown that $k = \delta_s/2\pi$. The motion algorithm is described below and the process is depicted in Fig. 4, with spiral branches labeled 1, 2, 3, ... n . The algorithm executes until the n^{th} branch of the spiral is completed or until a positive detection of biogas is made.

Algorithm: Follow spiral trajectory (ρ, δ_s, s, n)

1. Aim rover pointing system at retroreflector target; measure range, d_r , to the retroreflector
2. Turn to orient rover along rover-retroreflector pointing direction
3. Drive toward start of spiral trajectory, until $d_r = \rho$
4. Execute spiral survey
 - a. Orient rover tangent to spiral trajectory ($\perp$ to rover-retroreflector pointing direction)
 - b. Execute survey spiral branch i
 - i. Aim at retroreflector, acquire measurement(s)
 - ii. Move arc-length s to next M-node while increasing radius of curvature (Eq. 1)
 - iii. Repeat until spiral branch i is complete
 - c. Next branch, $i = i + 1$
 - d. Repeat until branch n complete or gas detected.

The rover can maneuver onto the spiral trajectory at the start of a survey using the same approach described above for circular trajectories. Recovering the spiral trajectory after deviating from it to avoid a hazard is facilitated by range and heading measurements to the retroreflector from which polar coordinates of the rover in the survey coordinate system can be determined. A close coordinate on the trajectory can be reached from off-trajectory by traversing inward or outward as necessary along the rover-retroreflector pointing direction to the proper spiral radius, ρ_s (recall Fig. 3).

For both the circular and spiral survey algorithms, the key parameters (ρ, δ, s, n) allow surveys to be configured in a flexible manner to achieve desired degrees of resolution and area coverage. Whether executed radially inward or outward, the surveys are constrained primarily by rover kinematic limitations, maximum range of the open-path spectrometer, and terrain topography in the survey region.

IV. MOBILE SURVEYING ARCHITECTURE

The surveying algorithms described above represent a fundamental capability that enables coverage of a survey region with discrete measurements of the near-surface atmosphere. To implement a more complete system that would operate robustly in a variety of field topographies additional behavioral capabilities are required. Among them is an obstacle or hazard avoidance behavior that keeps the rover safe in rough terrain and enables it to negotiate obstacles along the survey trajectory. Also required is a

behavior that would enable the rover to hone in on the location of any biogenic gas detected during a survey. Other special purpose capabilities could be imagined but these, including the survey trajectory following algorithms above, are a sufficient baseline set of behaviors. An architecture that appropriately uses the capabilities to achieve the overall objective of searching for, detecting, and finding biogenic gas sources is necessary to field a complete system for this task. In this section we discuss our approach in this regard.

A. Negotiating Obstacles on Survey Trajectories

A primitive obstacle avoidance behavior is used to negotiate obstacles when encountered. It employs a reactive sensor-based approach [11] using a laser range finder for obstacle detection. Qualitatively, the obstacle avoidance behavior steers the rover away from obstacles detected while following survey trajectories or navigating in general. It is activated by laser range inputs that fall within a short-range portion of the rover's sensor horizon. When no obstacles are detected in the immediate vicinity of the rover the behavior exhibits a tendency to move the rover in a forward direction; thus, when operating alone without system motion biases to follow a survey trajectory or navigate to a specific location the behavior displays a wandering activity.

B. Localizing Detected Biogenic Gas Sources

When an area of high biogas concentration is detected during a survey, its source can be localized as a result of rover motion towards the retroreflector along the detection azimuth. This capability is still under development but we describe the working concept as follows. Upon detection, the source is assumed to be between the rover and retroreflector and at least near the line-of-sight of the measurement. A stronger assumption is that the source lies not only along the detection azimuth but also within a separation distance, δ , to the previously traversed circle or spiral branch. It is further assumed that no terrain feature is blocking the rover's path except perhaps non-traversable obstacles of heights lower than the instrument laser beam height during the detection. As the rover moves toward the retroreflector along the detection azimuth, periodic measurements of gas concentration would be taken. Any isolated obstacles encountered along the detection azimuth would be avoided using the obstacle avoidance behavior while attempting to roughly follow the obstacle contour influenced by the motion bias to traverse a straight path. The activation of both behaviors continues until the obstacle is cleared and the rover can resume source localization.

The spectrometer is designed to maintain sensitivity to detectable gases at laser-retroreflector separation distances as short as 1 m. As such, the source of detected biogenic gas could be localized within considerable accuracy. Once a gas source is found, its location must be either recorded or marked to complete the localization task. If absolute position data were available, simple recording of the coordinates would be sufficient. In lieu of absolute position data, the location of the find could be marked by passive or active

markers, which would have to be carried by the rover during its search and deployed near the localized source.

C. Integrated Mobile Survey Approach

Fusion of behaviors is necessary for hazard-free trajectory following and particularly for dealing with survey trajectory recovery after avoiding isolated and extended obstacles. Currently, reactive trajectory recovery based on rover position knowledge is used and is built into the circular and spiral survey algorithms appropriate for the instrument system configuration (step 3 of either algorithm). Each motion behavior in the instrument mobility subsystem is part of a behavior hierarchy that operates using behavior fusion techniques described in earlier work [11]. The basic hierarchy of behaviors employed is shown in Fig. 5.

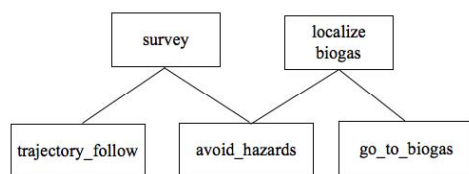


Fig. 5. Behavior hierarchy for mobile biogenic gas surveying.

A means to command the rover to execute a survey of a given area must be accounted for from a mission operations standpoint. The survey trajectory following algorithms are designed with this in mind; their specification using the four key parameters discussed earlier facilitates this. Mission operators would design a survey based on selection and specification of the four parameters. The radial distance from which to start a survey is designated by mission operators based on any known physical constraints in the local environment (e.g., terrain topography), and may be influenced by hypotheses about distributions of biogenic gas emissions in the area. Arc-increments, s , which separate the M-nodes are also specified by mission operators based on site-specific characteristics. For example, s may be chosen based on *a priori* information about the expected surficial size of emission sources in the area. Like s , the spiral branch separation distance, δ_s , may be specified based on information about the expected size of emission sources. It may also be governed by desired area coverage resolution or constrained by a need to complete a survey within available resources such as time, power, and data storage.

V. EXAMPLES

Computer simulations were used to develop and verify the survey algorithms to be later implemented on a Pioneer 2-AT commercial mobile robot. Fig. 6 shows an annotated screen shot of the simulation environment. It includes models of the physical rover platform, its laser range finder, a diode laser pointing subsystem on the simulated rover, a retroreflector and terrain, all modeled in 3-D using Webots (<http://www.cyberbotics.com>), a commercial mobile robot simulation software developed by Cyberbotics, Ltd.

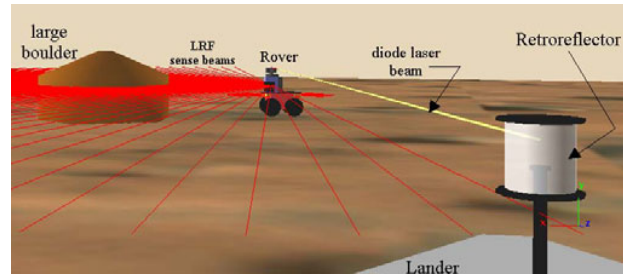


Fig. 6. 3-D simulation environment for survey task development.

Selection of one trajectory type over another for a given survey region can be based on a variety of factors including expected resource usage. Spiral surveys are generally more efficient in terms of the amount of traverse distance needed to cover an equivalent area [12]. A rover executing a concentric circular survey of n circles would need to traverse over $(4n-1)\delta_s$ meters more to cover the same area as it could with a spiral trajectory of n branches [12]. Despite the efficiency advantage of spirals, either trajectory type could be more feasible than the other in a given terrain topography.

A 2-D simulated execution of a survey trajectory of $n=3$ concentric circles separated by $\delta_c = 10$ m is shown in Fig. 7 in which the rover drives a linear distance of 397 m. Small rover icons are shown at M-nodes with dashed lines emulating the diode laser beam pointed at the retroreflector. Note that only every *third* rover icon and measurement beam is displayed in the figure to avoid obscuring the figure detail, thus the non-uniform beam spacing. To survey a roughly equivalent area using a spiral trajectory of $n = 3$ branches separated by $\delta_s = 10$ m the rover would only need to drive a linear distance of 283 m, or 29% less in this case; the spiral trajectory is shown in Fig. 8.

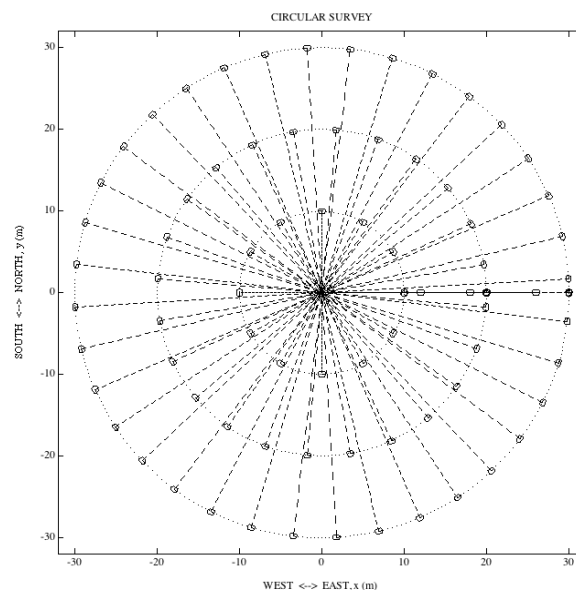


Fig. 7. 2-D concentric circular survey execution.

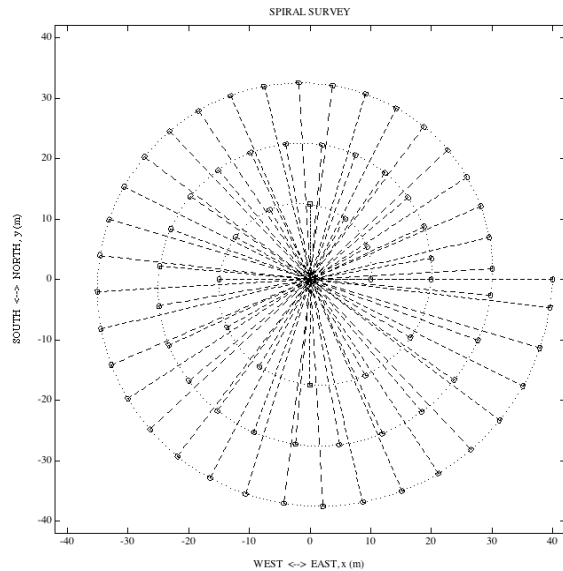


Fig. 8. 2-D spiral survey execution.

An example of trajectory following while avoiding obstacles encountered in the survey region is shown in Fig. 9 where the filled circles represent non-traversable obstacles. In this case, the commanded survey is one of concentric circles separated by $\Delta r = 4$ m, where only the first 2 circles traversed are shown. In Fig. 9, a rover icon is shown at every motion step, thus a uniform spacing of measurement beams is apparent. Note that no measurements are presently taken during obstacle avoidance, and that the obstacle in the bottom left of the figure is tall enough to actually obscure the measurements although the simulation does not show that effect. This result demonstrates the intended capability of the approach for hazard-free autonomous surveying of relatively open areas containing isolated obstacles.

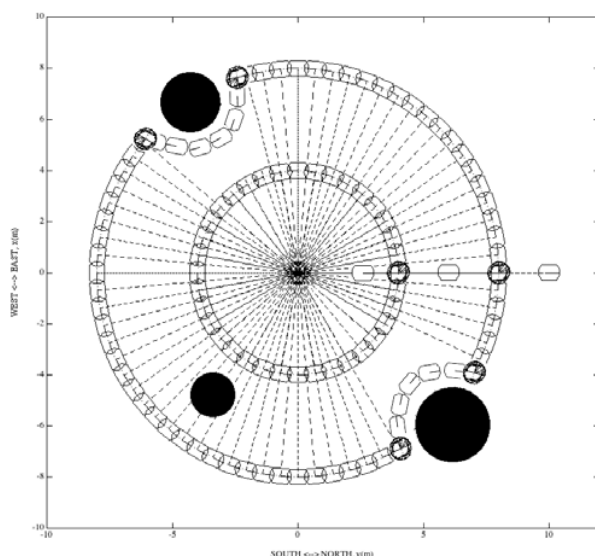


Fig. 9. Combined survey trajectory following and obstacle avoidance.

VI. CONCLUSIONS

A mobile surveying application was introduced involving remote-sensor-based area coverage using an open-path distributed spectrometer. Circular and spiral trajectories with *in situ* remote sensing is advocated as a wide-area coverage solution that obviates the need to traverse an entire survey area. Survey algorithm execution was demonstrated in simulation. While the underlying navigation technology needed to accomplish area coverage tasks is reasonably mature [13], the survey technology that would build on it to enable remote sensing surveys is not; this work represents necessary development in that area. As a next step, the survey trajectories will be integrated and tested on a physical mobile robot that will carry the spectrometer. Measurements will be acquired over the course of several outdoor field tests to survey Earth locations known to emit biogenic gases.

ACKNOWLEDGMENT

The research described in this paper was performed in part at the Jet Propulsion Laboratory, California Institute of Technology, under contract with the National Aeronautics and Space Administration, and at Harding University and the University of Arkansas at Little Rock with sponsorship of the NASA ASTID Program, Science Mission Directorate.

REFERENCES

- [1] G.T. Anderson, C. Sheesley et al, "A distributed diode laser spectrometer for mapping biogenic gases on the martian surface," Mars Atmos. Chem. & Astrobiology Workshop, Pasadena, CA, 2001.
- [2] T.D. Barfoot, G.M.T. D'Eleuterio, P. Annan, "Subsurface surveying by a rover equipped with ground-penetrating radar," IEEE/RSJ Intl. Conf. on Intell. Robots & Systems, Las Vegas, 2003, pp. 2541-2546.
- [3] J.D. Nicoud, "Vehicles and robots for humanitarian demining," *The Industrial Robot*, Vol. 24, No. 2, 1997, pp. 164-168.
- [4] D. Wettergreen, N. Cabrol, V. Baskaran, et al, "Second experiments in the robotic investigation of life in the Atacama Desert of Chile," 8th Intl. Symp. on AI, Robotics, & Automation in Space, Munich, 2005.
- [5] E. Garcia and P. Gonzalez de Santos, "Mobile-robot navigation with complete coverage of unstructured environments," *Robotics and Autonomous Systems*, Vol. 46, 2004, pp. 195-204.
- [6] G.T. Anderson, E.W. Tunstel and E.W. Wilson, "A robot system to search for signs of life on Mars," *IEEE Aerospace and Electronic Systems Magazine*, to appear 2007.
- [7] G.T. Anderson, R.R. Hashemi, E. Wilson, M. Clark, "Application of cooperative robots to search for water on Mars using distributed spectroscopy," 8th Intl. Symp. on Robotics & Appl., Maui, HI, 2000.
- [8] Y. Mei, Y.-H. Lu, Y.C. Hu, and C.S.G. Lee, "Energy-efficient motion planning for mobile robots," IEEE Intl. Conf. on Robotics and Automation, New Orleans, 2004, pp. 4344-4349.
- [9] S.C. Wong, L. Middleton, B.A. MacDonald, "Performance metrics for robot coverage tasks," Australasian Conf. on Robotics & Automation, Auckland, Nov. 2002, pp. 7-12.
- [10] S. Burlington and G. Dudek, "Spiral search as an efficient mobile robotic search technique," Technical Report, Ctr. for Intelligent Machines, McGill University, Montreal, Canada, Jan. 1999.
- [11] E. Tunstel, "Fuzzy-behavior synthesis, coordination, and evolution in an adaptive behavior hierarchy," in Saffiotti, A. and Drankov, D. (Eds.) *Fuzzy Logic Techniques for Autonomous Vehicle Navigation*, Physica-Verlag, Heidelberg; New York, 2001.
- [12] E. Tunstel, G. Anderson and E. Wilson, "Motion trajectories for wide-area surveying with a rover-based distributed spectrometer," 11th Intl. Symp. on Robotics & Applications, Budapest, Hungary, 2006.
- [13] H. Choset, K. Lynch, S. Hutchinson et al (Eds.), *Principles of robot motion: Theory, algorithms, and implementations*, MIT Press, Cambridge, MA, 2005.